

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI



A Project Report on

**LUNG CANCER DETECTION SYSTEM USING
CNN**

Submitted in partial fulfillment of the requirement for the award of
the degree of

MASTER OF COMPUTER APPLICATIONS

Under

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

By

PAWAN KUMAR K.P

4SO23MC068



**Department of Computer Applications
St Joseph Engineering College
Mangaluru-575028**

2025

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI



A Project Report On

**LUNG CANCER DETECTION SYSTEM USING
CNN**

Submitted in partial fulfillment of the requirement for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

Under

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

By

PAWAN KUMAR K.P

4SO23MC068

Under the Guidance of

Ms Sumangala N

Assistant Professor

Department of Computer Applications

St Joseph Engineering College

Mangaluru-575 028.



Department of Computer Applications

St Joseph Engineering College

Mangaluru-575028

2025

STJOSEPH ENGINEERING COLLEGE, MANGALURU
DEPARTMENT OF COMPUTER APPLICATIONS



CERTIFICATE

This is to certify that the project work titled

**LUNG CANCER DETECTION SYSTEM USING
CNN**

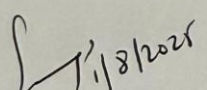
SUBMITTED BY

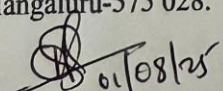
PAWAN KUMAR K.P

4SO23MC068

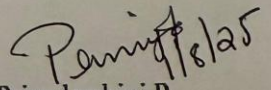
*In partial fulfillment of the requirements for award of degree of
Master of Computer Applications of Visvesvaraya Technological University,
is a Bonafide record of the work carried out*

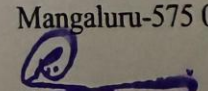
**During the academic year
2024-2025.**


Ms. Sumangala N
Assistant Professor
Department of Computer Applications
St Joseph Engineering College
Mangaluru-575 028.


Dr Hareesh B
HOD-MCA
St Joseph Engineering College
Mangaluru-575 028.

Examiner 1.


Ms. Priyadarshini P
Project Coordinator
Department of Computer
Applications
St Joseph Engineering College
Mangaluru-575 028


Dr Rio D'Souza
Principal
St Joseph Engineering
College
Mangaluru-575 028.

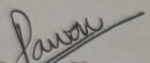
Examiner 2.

DECLARATION

I hereby declare that the entire work embodied in this dissertation has been carried out by me and no part of it has been submitted for any degree or diploma of any university previously.

Place: Mangaluru

Date: 31 July 2025


Name: Pawan Kumar

USN: 4SO23MC068

ACKNOWLEDGEMENT

It is my profound gratitude to thank one and all who have helped in bringing this project entitled “Lung Cancer Detection System Using CNN” up to the mark. Whatever I was able to put forward in terms of my work, is only due to few people from whom I learnt a lot.

Firstly, my sincere thanks to Rev. Fr Wilfred Prakash D’Souza, Director, SJEC, Fr Kenneth Rayner Crasta, Assistant Director, SJEC and Dr Rio D’Souza, Principal, SJEC, for their never-ending support and valuable suggestions. I wish to express my deep sense of gratitude to Mr. Hareesh B, HOD-MCA, for providing me the opportunity to take up this project. I would like to thank Ms. Sumangala N, Internal Guide & Assistant Professor-MCA, for her support and encouragement. I would also like to take this opportunity to convey my deep-felt gratitude to Ms. Priyadarshini P, Project Coordinator & Assistant Professor-MCA, who was there for us when required and I must say she has been the backbone of this Dissertation Report. I would also like to express my gratitude towards all the teaching and non-teaching staff of the Department of Computer Applications and to the other staff members of the college.

I extend my deepest gratitude to my beloved parents, whose unwavering love, sacrifices and constant encouragement have been the foundation of my journey. Their support—both emotional and moral—has been my greatest strength throughout the course of this project and my academic life.

I also wish to express my heartfelt thanks to my dear friends for their valuable suggestions, timely help and encouragement during various stages of the project.

Above all, I bow in reverence to God Almighty, whose infinite grace, guidance and blessings illuminated my path and gave me the strength and clarity to complete this project successfully. Without His divine presence and mercy, this work would not have been possible.

There are many others who have directly or indirectly helped me in my project work. The absence of their name does not show the absence of my gratitude for their support.

Pawan Kumar

31 July 2025

ABSTARCT

The rapid advancement of Artificial Intelligence in medical diagnostics has significantly improved the early detection of life-threatening diseases. This project presents a web-based lung cancer detection system designed using a Convolutional Neural Network (CNN) integrated with the Flask web framework. The core objective is to assist healthcare professionals by automating the classification of lung CT scan images into three categories: benign, malignant, and normal.

The proposed system employs a hybrid 2×2 CNN model that extracts spatial features through convolutional and pooling layers, followed by a dense classifier to determine the final prediction. The model is trained on a labelled CT scan dataset with appropriate preprocessing steps such as resizing, normalization, and augmentation. The classification output is further visualized using a graphical bar chart to enhance interpretability for end users.

The application includes three main user roles: Admin, Doctor, and Patient. Admins manage users and system data, doctors can view and assess medical predictions, and patients can upload their CT scan images to receive immediate feedback. The entire system runs on a Flask backend with MySQL database integration, offering a user-friendly interface for image uploads, prediction results, and report generation.

This system reduces diagnostic delay by providing automated predictions and promotes accessibility by offering remote usage. Future enhancements may include real-time image capture integration, multilingual support, and expansion of the model to detect additional respiratory diseases. By combining deep learning with web technology, this project aims to provide a scalable, efficient, and accurate decision-support tool for early lung cancer screening.

TABLE OF CONTENTS

1. Introduction	01
2. Literature Survey.....	02
3. Software Requirements Specification	04
4. System Design.....	13
5. Detailed Design.....	19
6. Implementation.....	31
7. Software Testing.....	35
8. Conclusion.....	41
9. Future Enhancements	42
Appendix A: References	43
Appendix B: Screenshots	44

INTRODUCTION

1.1 Project Description	01
-------------------------------	----

LITERATURE SURVEY

- 2.1 Existing and Proposed System02
 - 2.1.1 Existing System.....02
 - 2.1.2 Proposed System.....02
- 2.2 Feasibility Study.....02
- 2.3 Tools and Technologies Used.....03

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Introduction	04
3.1.1 Purpose	04
3.1.2 Document Conventions/Definitions and Abbreviations.....	04
3.1.3 Intended Audience and Reading Suggestions	05
3.1.4 Project Scope.....	05
3.1.5 Benefits	06
3.1.6 References	06
3.2 Overall Description	07
3.2.1 Identification of Pre-existing work.....	07
3.2.2 Product Perspective	07
3.2.3 Product Features	07
3.2.4 End User Characteristics	08
3.2.5 Operating Environment	08
3.2.6 Design and Implementation Constraints	08
3.2.7 Assumptions and Dependencies.....	08
3.3 Product Functionality	09
3.3.1 User Module	09
3.3.2 Admin Module	09
3.3.3 Doctor Module	09
3.4 External Interface Requirements	09
3.4.1 User Interfaces.....	09
3.4.2 Hardware Interfaces	10
3.4.3 Software Interfaces.....	10
3.4.4 Communication Interfaces	10

3.5 Non Functional Requirements	10
3.5.1 Performance Requirements	10
3.5.2 Safety Requirements.....	11
3.5.3 Software Quality Attributes.....	11
3.5.3.1 Reliability.....	11
3.5.3.2 Maintainability	11
3.5.3.3 Portability.....	11
3.5.3.4 Usability.....	11
3.5.3.5 Flexibility.....	11
3.6 Specific Requirements.....	12
3.6.1 Operating Environment	12
3.6.1.1 Hardware	12
3.6.1.2 Software	12
3.7 Delivery Plan.....	12

SYSTEM DESIGN

4.1 System Design.....	13
4.1.1 Introduction	13
4.1.2 Scope of The Project	13
4.1.3 Intended Audience	14
4.1.4 Context Flow Diagram	15
4.1.5 Admin DFD	16
4.1.6 Doctor DFD	17
4.1.7 Patient DFD	18

DETAILED DESIGN

5.1 Use Case Diagram.....	19
5.2 Sequence Diagrams	20
5.2.1 Admin Sequence Diagram	20
5.2.2 Doctor Sequence Diagram	21
5.2.3 Patient Sequence Diagram	22
5.3 Activity Diagram.....	23
5.3.1 Admin Activity Diagram	23
5.3.2 Doctor Activity Diagram.....	24
5.3.3 Patient Activity Diagram.....	25
5.4 Database Design.....	26
5.4.1 ER Diagram.....	29

IMPLEMENTATION

6.1 Introduction	30
6.2 Pseudo Codes	30
6.2.1 Pseudo Code for Register page	30
6.2.2 Pseudo Code for Login Page	31
6.2.3 Pseudo Code for Lung Cancer Detection.....	32
6.3 CNN Algorithm Details	33
6.4 Accuracy Assessment	33

SOFTWARE TESTING

- 7.1 Introduction.....34
- 7.2 Testing Object.....34
- 7.3 Unit Testing.....35
 - 7.3.1 Unit Test cases35
- 7.4 Integration Testing38
 - 7.4.1 Test Cases for Integration Testing39
 - 7.4.2 Other Cases39